After completion of the preparation materials, students should be able to:

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1. **Structure activity relationship:** Describe the structure-activity relationship of the two classes of local anesthetics.
2. **Pharmacokinetics:** Relate the influence of the PK properties on the clinical use of LAs.
3. **Properties:** Interpret how the drug pKa and tissue pH influence the rate of onset, potency, and duration of action of the ester-type and amide-type local anesthetics.
4. **Mechanism of action:** Explain the mechanism of action of local anesthetics with respect to state-dependent, use-dependent, and voltage-dependent block.
5. **Fiber susceptibility:** Differentiate the physiologic and structural factors that determine the susceptibility of nerve fibers to local anesthetic blockade.
6. **Clinical application:** Apply the fiber susceptibility factors to justify the therapeutic uses and routes of administration of the individual local anesthetics.
7. **Toxicity mechanisms:** Summarize the mechanisms underlying the major toxic effects associated with each local anesthetics.
8. **Vasoconstrictors:** Predict the effects of epinephrine on the duration of action, systemic absorption, and toxicity risk of local anesthetic therapy.
9. **Vasoconstrictor cautions:** Identify the mechanisms that influence cautions and contraindications of vasoconstrictor use.

Key points

Pharmacokinetic properties: ***Structural and physicochemical properties of the drugs*** influence protein binding, tissue distribution, onset and duration of action, metabolism, and toxicity.

Structure	aromatic (lipophilic) ring – ester or amide link – amine (hydrophilic) Weak bases with pKa between 7.5-9.0 (except benzocaine, 3.5)
Routes of administration	topical | infiltration | peripheral nerve block | I.V. regional nerve block | neuraxial blocks – spinal, epidural / caudal
Absorption	LAs are slowly absorbed from the site of injection. Doses are typically small. Extent and rate of absorption depend on dose | site | route | vascularity | drug properties Systemic absorption 1) terminates effect and 2) can cause adverse effects
Distribution	First pass extraction by lungs and other highly perfused organs, then sequestration in muscles and fat Bind tissue proteins and plasma proteins (alpha-1 acid glycoprotein)
Metabolism	Ester linked: Pseudocholinesterase to para-aminobenzoid acid (PABA) Amide linked: Hepatic oxidation / CYPs
Excretion	Urine, metabolites

Key points

t½	LAs are typically injected into confined spaces in tissues as a single bolus. ***Half-life becomes important when injected into the bloodstream*** (inadvertently or continuous infusion) for determining toxicity potential and the need for treatment.
Onset of action	Lower pKa (closer to physiologic pH ~7.4) → more uncharged base → faster onset
Potency	Greater lipid solubility → greater affinity for the lipid membrane → greater fraction of of the dose can enter the neurons → a lower dose is needed to produce an equivalent effect
Duration of action depends on how quickly the drug is cleared from the nerve.	**Vascularity at injection site** In areas of high blood flow, the LA is quickly removed → shorter duration of action • **Vasoconstriction** by epinephrine or other vasoconstrictor slows removal of the LA from the anesthetic site → systemic absorption is decreased and duration is increased. **Protein binding** LAs bind plasma proteins, tissue proteins and sodium channel proteins. Protein-bound drugs act as a reservoir, slowly releasing the drug. • **Lipid soluble drugs** tend to bind more tightly to tissue proteins and be retained in nerve membranes → longer duration of action.

Key points

Ester linked

Procaine	The prototype and no longer marketed. It has fast onset and short duration of activity.
Chloroprocaine	Very fast onset / short duration of action / rapid recovery → useful for outpatient procedures
Tetracaine	High potency / slow onset / long duration of action (1.5-3 hours) Spinal and topical use
Benzocaine	Topical only. It is almost entirely uncharged at physiologic pH Benzocaine readily penetrates mucous membranes to reach the nerves but is poorly absorbed through intact skin. Systemic toxicity: Methemoglobinemia
Cocaine	Topically for nasal and airway surgeries Its intrinsic vasoconstriction reduces bleeding and controls spasms
Role of Epinephrine	Vasoconstrictor typically used with short- and intermediate-acting LAs to prolong the duration of local or regional anesthesia (not obstetric) Inadvertent IV injection → hypertension, palpitations, cardiac arrhythmias

Key points

Amide linked

Lidocaine	Intermediate duration / broad use by several routes, IV antiarrhythmic
Mepivacaine	Intermediate duration / local and regional anesthesia / dental use
Prilocaine	Intermediate duration / methemoglobinemia due to o-toluidine metabolite Eutectic mixture of lidocaine + prilocaine → topical analgesic cream
Bupivacaine	High potency / High protein binding / Long duration / Many clinical applications Most frequently used LA worldwide for long-acting block and analgesia Highest potential for cardiotoxicity because of strong binding affinity
Ropivacaine	High potency but less than bupivacaine / High protein binding / Long duration Broad clinical use, including labor epidurals, spinal anesthesia for C-sections, ambulatory surgery. Ropivacaine differs from bupivacaine in sensory block with less intense motor block (faster return of function), reduced cardiotoxicity, lower CNS toxicity.
Role of Epinephrine	Vasoconstrictor typically used with short- and intermediate-acting LAs to prolong the duration of local or regional anesthesia (but not in obstetrics) Inadvertent IV injection → hypertension, palpitations, cardiac arrhythmias

Pharmacokinetics of systemically absorbed local anesthetic

<table>
<tr><td>Absorption
LAs are intended to remain at injection site</td><td colspan="2">LAs are typically slowly absorbed from the site of action.
Systemic absorption 1) terminates effect, 2) can cause adverse effects
Extent of systemic absorption is determined by dose, local vascularity, use of vasoconstrictor, physicochemical properties of drug</td></tr>
<tr><td rowspan="2">Protein binding plasma and tissue proteins</td><td colspan="2">Amide-linked local anesthetics are extensively (55% to 95%) bound to plasma alpha-1 acid glycoprotein.
Protein-bound drugs act as a reservoir, slowly releasing the drug.
• Highly lipid soluble drugs tend to bind more tightly to plasma and tissue proteins.</td></tr>
<tr><td>1) Distribute first to highly perfused tissues
2) Then to moderately perfused tissues</td><td rowspan="2">Potential for organ toxicity with high LA concentrations</td></tr>
<tr><td colspan="2">First-pass extraction: Highly perfused lungs sequester significant amounts of circulating LA before it reaches the arterial circulation. Brain, liver, heart, kidney also sequester LA.
Followed by distribution to muscles / fat (both have large capacity for storing LA)</td></tr>
<tr><td>Metabolism</td><td colspan="2">Ester-linked: Rapid by pseudocholinesterase to PABA (inactive)
Amide-linked: Hepatic CYPs</td></tr>
<tr><td>Excretion</td><td colspan="2">Metabolites are excreted in urine primarily
(very little unchanged drug)</td></tr>
</table>

Onset of action is related to pKa. Weak bases (pKa 7.5 to 9)

Comparing the ratios of non-protonated to protonated forms of LAs with different pKa values at the same pH (~7.4):

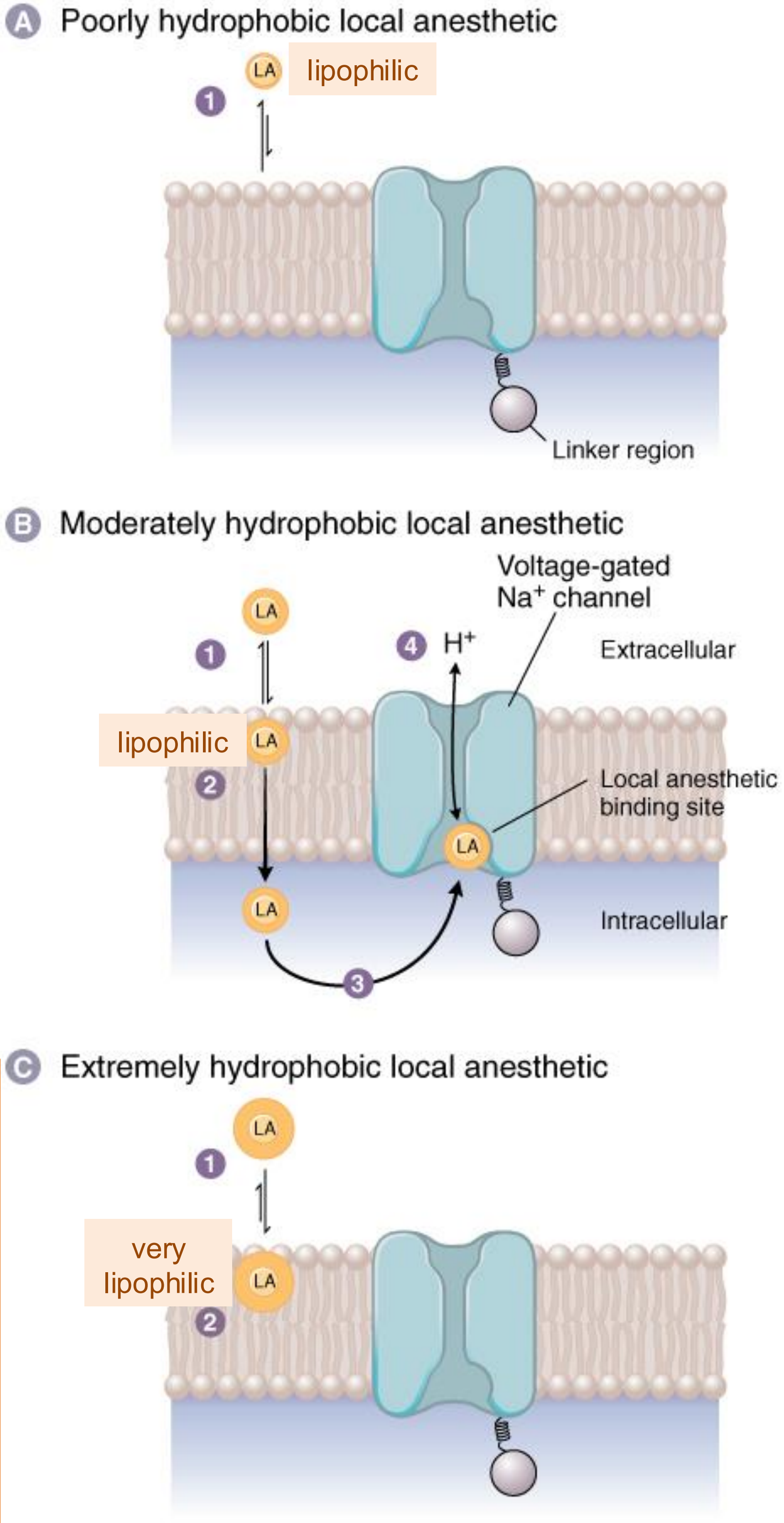
$$LA + H^{+} \leftrightarrows LAH^{+}$$

$$pK_a = pH - \log[B]/[BH^{+}]$$

- The LA with the pKa closer to physiological pH will have a greater fraction of the drug in the uncharged form than one with a higher pKa.
- The uncharged form diffuses through the membrane.

Onset of action:

Therefore, a higher concentration of the LA with the pKa closer to pH 7.4 is delivered to the site of action → more rapid onset of action.



Practice point:

Infected tissue is more acidic (lower extracellular pH).

A greater fraction of the drug is in the ionized (LAH^{+}) form → a lower concentration of uncharged base.

Thus, local anesthetics are less effective when injected into infected tissue.

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Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy, 4e, 2017
Figure 12-5

Physicochemical Characteristics

ESTER-LINKED	Relative POTENCY	DURATION OF ACTION
✳Procaine	1	Short
Chloroprocaine	~2	Short
✳Cocaine	2	Medium
✳Tetracaine	16	Long
✳Benzocaine	<1	Topical only
AMIDE-LINKED		**DURATION**
✳Lidocaine	4	Medium
Mepivacaine	2	Medium
✳Prilocaine	3	Medium
✳Bupivicaine	16	Long
✳Ropivacaine	slightly <16	Long
Articaine	not listed	Medium

Stereochemistry: *S*-enantiomer is less toxic than the R-enantiomer (in general)

Potency is related to lipid solubility.

Highly lipid soluble LAs easily diffuse through neuron membranes.

greater lipid solubility

↓

greater affinity for the lipid membrane

↓

a greater fraction of molecules in the dose can enter the neurons

↓

a lower dose is needed
to produce an equivalent effect

Duration of action

Determined by how quickly the drug is cleared from the nerve, which depends on local blood flow and protein binding, rather than metabolism.

Vascularity at injection site

- In areas of high blood flow, the LA is quickly removed → shorter duration of action

  Vasoconstriction by epinephrine or other vasoconstrictor slows removal of the LA from the anesthetic site → systemic absorption is decreased and duration is increased.

Protein binding

- LAs bind plasma and tissue proteins and sodium channel proteins.
- Protein-bound drugs act as a reservoir, slowly releasing the drug.

  Highly lipid soluble drugs tend to bind more tightly to plasma and tissue proteins and be retained in nerve membranes → longer duration of action

Ester-linked LAs have a shorter duration of action because they are rapidly metabolized by pseudocholinesterase.

LA mechanism of action and the terms that describe it

The nonionized form of the LA passes through the nerve membrane and re-equilibrates in the cytosol into its ionized and nonionized forms.

The *ionized* (LAH+) *form* of the LA preferentially binds the sodium channel in the open and inactivated state by binding a site in the inner vestibule → stabilizes the fast inactivation state, accumulates block during repetitive firing, and binds more strongly at depolarized potentials, with minimal effect on normal low frequency signaling.

- **State-dependent block:**

  Channels in the open and inactivated state have a much higher affinity for the local anesthetic than do channels in the closed resting state. They enhance the fast inactivation of the channels. Recovery from drug-induced block is 10 to 1000 times slower than recovery of channels from normal inactivation.

- **Use-dependent block:**

  Rapidly firing nerves are preferentially blocked by LAs. Blockade by local anesthetics is more marked at higher frequencies of depolarization (accumulates in rapidly firing axons) than in resting fibers.

- **Voltage-dependent block:**

  Voltage changes cause changes in the conformation of the inner pore during channel opening and activation (voltage-dependent gating). More positive membrane potentials (as in the depolarized state) increase the degree of anesthetic block.

Check your knowledge

How do the pharmacokinetic properties of local anesthetics influence clinical use?

How is the pKa of a LA related to the onset of action?

How is the potency of a LA related to its lipid solubility?

Check your knowledge

What factors determine the duration of action of a local anesthetic?

What are the mechanisms?

Local Anesthetics Toxicity Mechanisms and Effects

Local anesthetic systemic toxicity (LAST) – CNS and cardiotoxicity
Transient neurologic symptoms (TNS)
Cauda equina syndrome
Methemoglobinemia
Hypersensitivity

LAs are intended to remain at the site of injection.

- Systemic absorption of toxic concentrations of drug can occur by inadvertent vascular administration or continuous infusion.
- The rate and extent of systemic absorption are determined by dose, site of administration, local vascularity, presence of vasoconstrictor, first pass extraction, sequestration in muscle and fat, plasma protein binding, and metabolism.

LA systemic toxicity (LAST): CNS and CV toxicities / Can occur with all LAs

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LAST may be fatal. The danger is proportional to LA concentration in plasma.

***Clinical harbinger:* CNS symptoms typically appear before cardiovascular collapse → an *early warning sign* of impending cardiotoxicity.**

CNS Toxicity	Cardiovascular toxicity
Early excitation: Perioral numbness, metallic taste, dizziness, nystagmus, tinnitus Sympathetic outflow is transiently increased → then ↓ sympathetic outflow	↑Sympathetic outflow → Initial ↑BP and tachycardia → then vasodilation / hypotension, bradycardia **Direct cardiovascular Na+ block:** Hypotension, bradycardia, depression of conduction velocity → indirect ↓ Ca2+ influx → ↓ myocardial contractility Mitochondrial dysfunction → ↓ energy output
Followed by CNS depression: Agitation, muscle twitching → CNS and myocardial depression → death by respiratory depression	**Followed by:** ventricular arrhythmias, **CARDIOVASCULAR COLLAPSE**

Treatment: Intralipid therapy, seizure control, and cardiopulmonary resuscitation

✳**Lipid rescue:** Proposed mechanisms: I.V. Intralipid emulsion creates an expanded lipid compartment that extracts LA from the blood (lipid sink) and redistributes (lipid shuttle) it to the liver (detoxification) and storage organs. Lipid therapy supplies long-chain fatty acids for ATP production.

Lidocaine spinal anesthesia: Transient neurologic symptoms (TNS)

- Pain or burning, prickly, achy sensation (dysesthesia) radiating from the lower back/gluteal region to the posterior thighs.
- Pain can be mild to severe. TNS is not associated with sensory loss, motor weakness, or bowel and bladder dysfunction.
- Reversible: Symptoms usually resolve after several days without sequelae.

Serious and permanent complications after spinal anesthesia are very rare.

- Mepivacaine: occurs / lower incidence
- Not seen with the other LAs.

Lidocaine spinal administration: Cauda equina syndrome (very rare)

- Irreversible. Compression of the nerve root bundle extending from the end of the spinal cord → saddle anesthesia/paresthesia, urinary / fecal incontinence and lower extremity weakness

Methemoglobinemia

- **Prilocaine** → toxic metabolite o-toluidine → methemoglobinemia → cyanosis → death potentially
- **Benzocaine** (pKa 3.5) aerosol spray→ well absorbed through mucous membranes during bronchoscopy, endoscopy, transesophageal echocardiography → absorption → toxic N-hydroxy aromatic amines → methemoglobinemia
- Avoid benzocaine in teething infants and children's mouth sores.

Other Effects

Hypersensitivity:

- **Ester LAs:**

Metabolized to *p*-aminobenzoic acid (PABA), which has been implicated in hypersensitivity reactions.

Contraindication to all ester LAs.

- **Amide LAs:**

Very low incidence of hypersensitivity reactions.

A patient allergic to one amide LA may be given a different amide.

Pregnancy:

epidural or spinal administration depending on the drug and clinical use:

labor or vaginal or cesarean delivery

Local anesthetics commonly used in labor and delivery:

- Bupivacaine
- Ropivacaine
- Tetracaine
- Lidocaine

Caution: Epinephrine should be avoided in obstetrical patients.

Vasoconstriction can compromise blood flow to the uterus and placenta, and risk maternal hypertension, tachycardia, and arrhythmias if absorbed systemically.

Check your knowledge

- What are the manifestations of LAST?

What is the reversal agent for LAST?

Which LA is associated with the highest risk of cardiotoxicity?

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Check your knowledge

What are the cause, manifestations, and prognosis of TNS?

What are the cause, manifestations of cauda equina syndrome, and potential prognosis of it?

What are the causes and effects of methemoglobinemia?

Which LAs have the highest incidence of hypersensitivity reactions?

Ester-linked local anesthetics: Pharmacokinetics

Procaine, Chloroprocaine, Tetracaine, and Benzocaine

Pseudocholinesterase: Rapid hydrolysis (<1 min) → *p*-aminobenzoic acid (PABA)
→ Short plasma half-lives
→ Reduced / absent plasma cholinesterase → drug accumulation → toxicity risk
→ PABA may cause a hypersensitivity reaction in PABA-allergic patients

- Note: Spinal fluid contains little or no esterase → anesthetic effect persists until agent has been absorbed into the circulation

Excretion: Urine, PABA and ionized drug

Cocaine (natural benzoic acid ester of *Erythroxylum coca*)

Metabolized by pseudocholinesterase and hepatic esterases → toxic N-hydroxy aromatic amines

Ester-linked local anesthetics

Drug	Clinical use	Properties	Toxicities
Procaine The prototype Manufacture discontinued	Subcutaneous infiltration for minor skin and dental procedures	Low potency Rapid onset Short-acting	All ester-linked LAs: Hypersensitivity reactions in PABA-allergic patients
Chloroprocaine	Infiltration Peripheral nerve block Spinal anesthesia	Low potency Very rapid onset Short-acting	As for procaine
Tetracaine	Spinal anesthesia	Highly lipophilic High potency Long-acting (2-3 h)	Significant toxicity when used for high-volume peripheral block
Benzocaine	Topical only (rapid absorption from mucous membranes but poorly from skin) Various forms, including aerosol spray	Low pKa (3.5) $\rightarrow$ exists as nonionized base at physiologic pH	Methemoglobinemia Aerosol spray for bronchoscopy, endoscopy, transesophageal echocardiograph
Cocaine	Topical for nose, mouth, pharynx procedures (highly vascular tissues)	Intrinsic vasoconstriction Rapid onset Short-acting	Abuse potential (dopamine, serotonin, norepinephrine reuptake inhibitor)

Amide-linked local anesthetics: Properties and Therapeutic Uses

Clearance is highly dependent on hepatic blood flow, hepatic extraction, enzyme function.

Drug	Properties	Clinical uses
Lidocaine	Rapid onset, moderate potency, moderate duration Hepatic CYP1A2,3A4 metabolism to active xylidide metabolites	Local, regional, neuraxial anesthesia Topical derm products, eye, ear, mouth, transdermal patch (OTC and Rx)
Prilocaine	Rapid onset, moderate potency, moderate duration Hydrolysis, hepatic and renal amidases to ***o*-toluidine** (oxidizing metabolite → methemoglobin; dose-depen.) Metabolite excreted in urine	Lidocaine + Prilocaine topical cream (eutectic mixture) The only use currently available in US
Mepivacaine	Rapid onset, moderate potency, moderate duration Hepatic metabolism; metabolites excreted in urine	Dental anesthesia Local, regional, neuraxial anesthesia
Bupivacaine	Rapid onset, high potency, long-acting Hepatic CYPs; metabolites excreted in urine	Local, regional, neuraxial anesthesia Low dose: Pain management Higher dose: Sensory and motor block
Ropivacaine	Rapid onset, high potency, long-acting Hepatic CYP1A2 and other CYPs; urine / metabolites	Local, regional, neuraxial anesthesia Pain management: post-surg / obstetrics
Articaine	Plasma carboxyesterase to inactive metabolite; metabolite excreted in urine	Dental anesthesia (formulated with epinephrine)

Amide-linked local anesthetics: Toxicities Reminder

Drug	Toxicities
LAST	All LAs can cause local anesthetic systemic toxicity – neurotoxicity and cardiotoxicity – in clinical concentrations. • Degree of toxicity is dose-dependent and drug-dependent • Lipid rescue: Administration of intravenous lipid emulsion 20%.
Bupivacaine is the most cardiotoxic of the LAs. CNS and CV toxicity occur almost simultaneously.	
Lidocaine	• Transient neurological symptoms (TNS) – Characterized by reversible back pain with radiation to buttocks or legs following spinal anesthesia. – High incidence (14-37%) with spinal lidocaine – Resolves in a few days usually without sequelae – Mepivacaine: occurs but lower incidence • Cauda equina syndrome: Irreversible nerve injury (very rare) following spinal anesthesia: saddle anesthesia or paresthesia, urinary and/or fecal incontinence, lower extremity weakness
Prilocaine	Methemoglobinemia due to o-toluidine (oxidizer) → cyanosis in susceptible patients

Structure and classification:

- A lipophilic aromatic ring determines lipid solubility and potency.
- An intermediate ester or amide chain classifies the drugs and determines its metabolic pathway.
- A hydrophilic amine group affects the drug's pKa and water solubility.
- Esters: procaine, chloroprocaine, tetracaine, cocaine, benzocaine
- Amides: lidocaine, mepivacaine, prilocaine, bupivacaine, ropivacaine

Pharmacokinetics

- pKa and onset: pKa determines the ratio of ionized to nonionized forms at physiological pH. The nonionized form is lipid-soluble and can cross the nerve cell membrane. Once inside the cell, it re-ionizes. LAH+ binds the target inside the Na+ channel from the cytoplasmic side.
- Lipid solubility and potency: Higher lipid solubility correlates with increased potency and faster onset of action.
- Protein binding and distribution: High protein binding in the plasma and tissues correlates with longer duration of action.
- Vasoconstrictors: Agents like epinephrine are often added to local anesthetics to cause vasoconstriction, which limits systemic absorption, prolongs the local effect, and reduces the risk of systemic toxicity.

Mechanism of action

- Target: primarily voltage-gated Na+ channels on the neuronal membrane
- Blockade – state-, use-, voltage-dependent: LAs preferentially bind the open and inactivated state of the Na+ channel. LAs enhance the fast inactivate state (prolongs block), accumulate block in frequently firing neurons, and bind more strongly at depolarized potentials.
- Differential block: Small myelinated and unmyelinated nerve fibers (Aδ and C fibers) that transmit pain and temperature are generally more susceptible to block than larger myelinated motor fibers, which can result in a differential loss of sensation. Sympathetic nerve fibers are most sensitive.

Toxicity

- Systemic toxicity is dose-dependent. It can occur if large amounts of LA enter the bloodstream.
- CNS: Initial symptoms may include circumoral numbness, tinnitus, dizziness, progressing to muscle twitching, CNS depression, seizures, and ultimately respiratory depression. CNS symptoms typically appear before cardiovascular effects.
- Cardiovascular effects: At very high concentrations, LAs can cause hypotension, bradycardia, depressed cardiac function leading to arrhythmias and eventually cardiovascular collapse.
- Bupivacaine is the most cardiotoxic LA. It tightly binds Na+ channels. CNS and CV toxicity occur almost simultaneously.
- TNS, caudal equina syndrome: Lidocaine spinal anesthesia
- Methemoglobinemia: toxic metabolites of prilocaine and benzocaine

Check your knowledge

How do the pharmacokinetic properties of local anesthetics influence clinical use?

- ***Structural and physicochemical properties of the drugs*** influence protein binding, tissue distribution, onset and duration of action, metabolism, and toxicity.
- Ester-linked LAs are rapidly metabolized by pseudocholinesterase so tend to have a shorter duration.
- Amide-linked LAs are metabolized by the liver so tend to have a longer duration.

How is the pKa of a LA related to the onset of action?

- The LA with the pKa closest to physiological pH will have a greater proportion of the molecule in the uncharged form (compared to a drug with a higher pKa). The uncharged form diffuses through the membrane. Therefore, a higher concentration of the LA is delivered to the site of action → more rapid onset.

How is the potency of a LA related to its lipid solubility?

- Highly lipid soluble LAs easily diffuse through neuron membranes, whereas penetration by less lipid soluble LAs is less efficient. Highly lipid soluble LAs have greater affinity for the nerve cell membrane (and surrounding tissues). Thus, a lower dose is needed to produce an equivalent effect. Furthermore, higher lipid soluble LAs have also have a longer duration of action.

Check your knowledge

What factors determine the duration of action of a local anesthetic?

- Duration of action is primarily determined by how quickly the drug is cleared from the nerve, which depends on local blood flow and protein binding, rather than metabolism.

What are the mechanisms?

- **Vascularity:** High vascularity (high blood flow) at the local site → faster washout → shorter duration
  - ➢ Vasoconstriction by epinephrine or other vasoconstrictor slows removal of the LA from the anesthetic site → systemic absorption is decreased and duration is increased.
- **Protein binding:** Binding to tissue proteins acts as a reservoir that slowly releases the drug.
  - ➢ Highly lipid soluble LAs tend to bind more tightly to tissue proteins and are retained in nerve membranes → longer duration of action

Check your knowledge

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- What are the manifestations of LAST?

CNS:

- Sympathetic NS: Sympathetic outflow is transiently increased → then ↓ sympathetic outflow
- Early symptoms: perioral numbness, metallic taste, tinnitus, nystagmus
- Later CNS symptoms: agitation, muscle twitching → CNS and myocardial depression → death by respiratory depression

Cardiovascular:

- Sympathetic → Initial ↑BP and tachycardia → then vasodilation / hypotension, bradycardia

Direct cardiovascular Na+ block:

- Hypotension, bradycardia
- depression of conduction velocity → indirect ↓ in Ca2+ influx → ↓ myocardial contractility

Mitochondrial dysfunction → ↓ energy output

What is the LAST reversal agent?

- LAST may be fatal. Lipid rescue: IV lipid emulsion administration

Which LA is associated with the highest risk of cardiotoxicity?

- Bupivacaine rapidly and strongly binds sodium channels and dissociates slowly (fast in, slow out) → ***Bupivacaine is the most cardiotoxic LA.***
- ***CNS and CV toxicity occur almost simultaneously.***

Check your knowledge

What are the cause, manifestations, and prognosis of TNS?

- Lidocaine spinal anesthesia may cause transient neurological symptoms: pain, burning, prickly, achy sensations (dysthesia), which is not associated with sensory loss, motor weakness, or bowel and bladder dysfunction
- Reversible: Usually resolves within several days.
- Lower incidence with mepivacaine.

What are the cause, manifestations of cauda equina syndrome, and potential prognosis of it?

- Lidocaine subarachnoid administration can very rarely lead to irreversible compression of the nerve root bundle extending from the end of the spinal cord, which can lead to urinary / fecal incontinence and lower extremity weakness

What are the causes and effects of methemoglobinemia?

- Prilocaine toxic metabolite o-toluidine; benzocaine (spray) toxic metabolite N-hydroxy aromatic amines. Can lead to cyanosis.

Which LAs have the highest incidence of hypersensitivity reactions?

- Ester-linked LAs are rapidly metabolized to PABA, which is associated with hypersensitivity reactions.
- Amide-linked LAs have a low potential for allergic reaction.